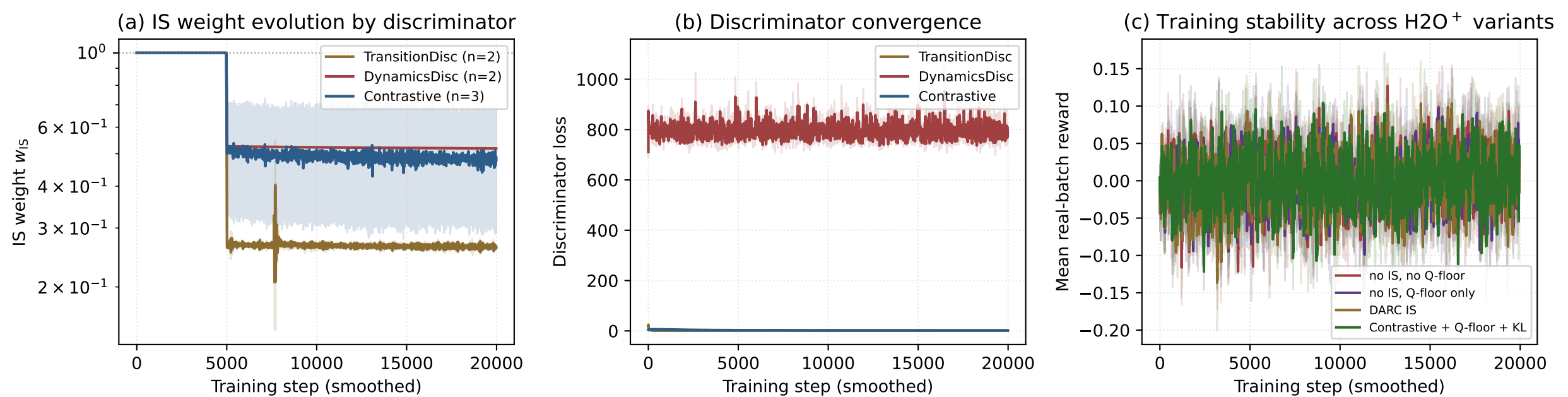

(a) IS weight evolution by discriminator
IS weight w_{IS}
10^0
6×10^{-1}
4×10^{-1}
3×10^{-1}
2×10^{-1}
TransitionDisc (n=2)
DynamicsDisc (n=2)
Contrastive (n=3)
0
5000
10000
15000
20000
Training step (smoothed)
(b) Discriminator convergence
Discriminator loss
1000
800
600
400
200
0
TransitionDisc
DynamicsDisc
Contrastive
0
5000
10000
15000
20000
Training step (smoothed)
(c) Training stability across H2O$^+$ variants
Mean real-batch reward
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
no IS, no Q-floor
no IS, Q-floor only
DARC IS
Contrastive + Q-floor + KL
0
5000
10000
15000
20000
Training step (smoothed)